

Insight Timer: Bringing a feature-rich meditation app to 15M+ users worldwide with Google Cloud

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InsightTimer

Insight Timer improved scalability, reliability and security by moving applications to Google Cloud and using Cloud Vision AI to safeguard online community.

About Insight Timer

Insight Timer is a meditation and mental health app that caters to people from all around the world. Founded in 2009, it operates out of Australia and currently has more than 15 million users. Its mission is to provide everyone access to a free daily meditation practice and facilitate a community of mental health professionals.

Industries: Healthcare

Location: Australia

Google Cloud results

- Reduces DevOps time with the automation of Firebase
- Supports reduced spend on infrastructure maintenance and enables investment in front end to improve app interface and features
- Improves app security with Firebase Authentication to protect millions of users' personal data
- Helps to detect spam messages with Cloud Vision AI, ensuring a positive

Lower
infrastructure
cost
a new
50%

Google Cloud (<https://cloud.google.com/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>),

App Engine (<https://cloud.google.com/appengine/>)

Cloud Speech to Text (<https://cloud.google.com/speech-to-text/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Firestore

(<https://cloud.google.com/firestore/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>),

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Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Cloud Speech to Text

(<https://cloud.google.com/speech-to-text/>)

Cloud Speech to Text

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud Vision API (<https://cloud.google.com/vision/>)

Firebase (<https://firebase.google.com/>)

Google Maps Platform

(<https://cloud.google.com/maps-platform/>)

Google Places API

(<https://cloud.google.com/maps-platform/places/>)

Recommendations AI

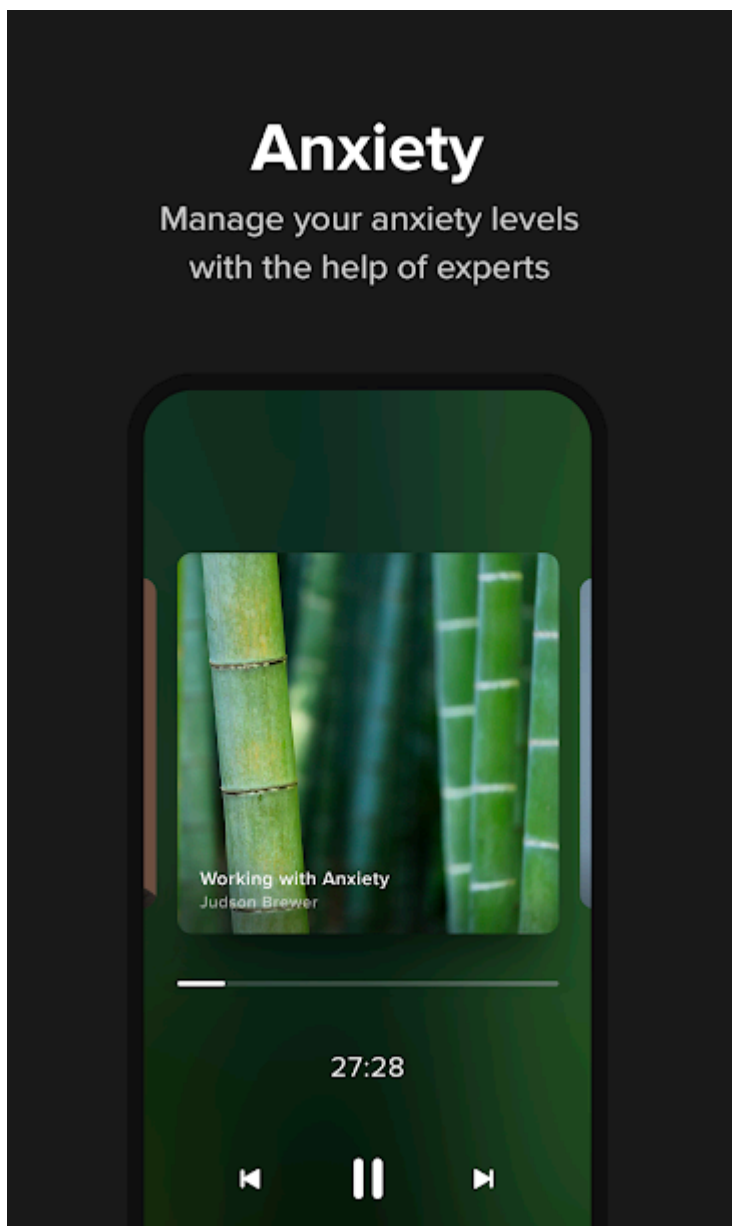
(<https://cloud.google.com/recommendations/>)

experience for
Insight Timer's
online
community

All over the world, 200 to 500 million people
meditate every day

(<https://www.thegoodbody.com/meditation-statistics/>),
with 76% seeing it as a good way to improve their
general wellness. Bringing this ancient Indian
practice to the modern world, mobile app Insight
Timer (<https://insighttimer.com/>) was founded 10 years
ago to give people access to free meditation. The
app provides a range of meditation music and helps
document the user's personal meditation journey.
For those who are more serious about the practice,
the app also provides mental health and meditation
classes conducted by certified teachers.

"Our goal has always been to provide free meditation
for everyone, as well as to provide income for
meditation teachers, so they can spend more time
teaching," says Cyrus Patel, Chief Technology Officer
at Insight Timer. The app works on a freemium
model, which means that it's free to use but people
can choose to pay a subscription fee if they want to
join the classes. Teachers are paid 50% of the fees if
a student chooses their class.



Today, Insight Timer is one of the largest online mental health communities, with more than 15 million users, including teachers, spiritual leaders, and meditators from around the world. Positioning itself as a platform that's available around the clock for meditation teachers and students, Insight Timer turned to Google Cloud (<https://cloud.google.com/>) to improve its infrastructure and app capabilities so it could support growing demand.

"Millions of people around the world rely on us to remain highly available, and we realized that the

stability, speed, and security of Google Cloud would help us achieve that goal," says Cyrus.

"We don't have to worry about downtime anymore since deploying Google Cloud. We can be confident that our app will work fine even during spikes in our traffic because Firebase is scalable and takes care of all the troubleshooting."

*—Cyrus Patel, Chief
Technology Officer,
Insight Timer*

A smooth move to Google Cloud to manage traffic spikes with confidence

Many people choose to meditate when they wake up in the morning, and that's when Insight Timer sees spikes in the number of people using its app. "We have a lot of users based in the United States," says Cyrus. "So, when everyone wakes up in the morning and starts meditating in New York, for example, we see a peak in app traffic. Meanwhile, it's nighttime for our engineers in Sydney and India. They're on the other side of it all, busy making sure that the app is able to support this spike in traffic," says Cyrus.

He explains that the team previously had to maintain its servers manually because the system setup heavily relied on DevOps. Because of this, engineers often had to be up at odd hours of the night to troubleshoot any issues and ensure minimal downtime. With Google Cloud, these things are no longer a concern.

"We don't have to worry about downtime anymore since migrating to Google Cloud. We can be confident that our app will work fine even during spikes in our traffic because Firestore (<https://firebase.google.com/>) is scalable, and takes care of all the troubleshooting," says Cyrus, adding that Google Cloud serverless components such as Pub/Sub (/pubsub) and App Engine (/appengine) also help to manage DevOps automatically.

So far, Insight Timer has moved about 70% of its app content to Google Cloud, in a smooth process that

Cyrus likens to water being pumped from one barrel to another. “We were able to migrate without disrupting the user experience because we did everything methodically and slowly,” he explains. “Over time, the new ‘water barrel’ on Google Cloud is getting fuller and fuller, as we empty the old barrel and it gets less traffic.”

That process began with migrating content such as courses and guided meditations, while testing the live migration at each point of the rollout. Insight Timer also made sure to maintain backwards compatibility by keeping both back ends running simultaneously, so that users who hadn’t upgraded their app yet would still be able to access everything they needed.

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The next step was to migrate user information such as bookmarks saved, progress stats, and group messages, a more complicated process due to Insight Timer's massive user base. However, the company doesn't foresee any issues and expects to complete its migration by the end of the year.

Because so many of its processes are now automated, Insight Timer has been able to reduce its operational costs, which means that the cost of infrastructure needed to support a new user is now 50% lower.

"The efficiency and reliability of Firebase has helped us to reduce cost by 50% per user since we began the migration process," says Cyrus. "Focusing on serverless products like Google Cloud means that we don't need a team dedicated to DevOps and we can invest in hiring more frontend engineers to focus on improving the user interface of the app."

Insight Timer has also been operating a lot more smoothly, with less interruption for its customers. "We can tell from our ratings on the app store that customers are generally happier with our app, as our rating is now at a 4.9 out of five stars, up from our previous rating of 4.6."

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—Cyrus Patel, Chief
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Insight Timer

Enhancing app security and personalization

More than just being a meditation app and learning tool, Insight Timer is also a social network that allows its users to chat with one another. This makes security an important aspect, as the app also contains a lot of user data such as chat history. "By using [Firebase Authentication](#)

(<https://firebase.google.com/docs/auth>), we don't have to build our own security system because the Google team has built it for us using the most trusted authentication standards that include strong encryption, among other things," says Cyrus. "When there's a new authentication system that's available, such as when Apple or Android added a new layer of security when signing in, for example, Firebase Authentication will make sure that we have the most updated systems in place to manage the change."

Because Insight Timer is also a social network where people can make friends and send messages to each other, Cyrus says it plans to use more machine learning abilities to detect users that are causing spam. "Currently, we already use [Cloud Vision](#) (/vision) to scan the images that are being uploaded so we can monitor our system for spam and quickly detect messages and pictures that shouldn't be there, keeping a safe environment."

Insight Timer also uses Google Cloud APIs for machine learning and location tracking to learn more about its users. "We use [BigQuery](#) (/bigquery) to collect data so that we can provide better recommendations by understanding what customers want to see. We also use the [Places API](#) (<https://developers.google.com/places/web-service/intro>) from [Google Maps Platform](#) (/maps-platform) to help them discover and connect with people around their vicinity," says Cyrus.

Future plans with machine learning

Cyrus explains that being on Firebase has already opened doors to a much quicker development cycle. Without being hampered by the nitty-gritty backend technicalities, the team at Insight Timer can focus on improving its app and experiment with new functions. "The [Filestore](#) (/filestore) and [Cloud Storage](#) (/storage) client side SDKs have allowed our iOS, Android, and Web teams to rapidly prototype new features and products without being blocked by requiring traditional backend APIs," says Cyrus. "This means that our client team can work autonomously and at a much quicker speed, because they already have the necessary database access."

The app currently supports 44 different languages including Korean, Danish, and Arabic, and includes content that caters to a variety of practices, from focused to movement meditation. And according to Cyrus, Insight Timer has exciting plans to expand its offering further still. "We want to do a lot more work on translation, for both audio and text," he shares.

“For this, we’re definitely looking to tap into the machine learning capabilities of Google Cloud, such as Speech-to-Text (/speech-to-text) API. We also want to categorize our meditation content into multiple themes and topics in order to build a good recommendation engine for our users, and we’ll use Speech-to-Text API to extract and categorize the spoken parts.”

As a company with big ambitions, Cyrus says, "Google Cloud is an incredible product. The meditation app space is a competitive one, but we’re scaling at a much faster pace than ever before, thanks to the serverless design of Firebase. I’m excited and look forward to growing our business even further with Google Cloud as our partner."

Music

Relaxing music and nature
sounds for drifting away

